

# Madhav Anand Menon

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## EDUCATION

**University of Illinois Urbana-Champaign**

Expected May 2028

**B.S. Computer Science and Physics; Minor in Mathematics (James Scholar)**

**GPA: 3.97/4.0**

- Yee Seung Ng Award; Harold and Ruth Hayward Scholarship; Dean's List; Tau Beta Pi Honours Society
- Relevant Coursework: Honours Data Structures and Algorithms, Computer Architecture, Honours Linear Algebra (Upcoming: Operating Systems, Numerical Methods)

## EXPERIENCE

**Valeo**

Jun. 2026 - Aug. 2026

Software Engineer Intern (Autonomous Vehicles Division)

Chennai, TN, India

- Implementing a Complementary Filter for the BHI 360 IMU sensor to characterise vehicle orientations under dynamic manoeuvres

**Disruption Lab**

Feb. 2026 - Present

Software Engineer

Champaign, IL, USA

- Built a 5-stage Canvas-to-PostgreSQL RAG pipeline supporting 3 file formats (PDF/DOCX/TXT) with 1024-dim pgvector embeddings and sliding-window chunking via AWS Bedrock, reducing retrieval latency by 35%
- Designed a skill extractor with NACE framework alignment, returning deduplicated skill taxonomies across 100+ courses per API call, reducing manual tagging effort by 98%
- Shipped a FastAPI backend with stateless HMAC-SHA256 authentication (8-hr TTL, zero external dependencies) and dual-mode search (full-text PostgreSQL and pgvector cosine similarity) supporting 120+ concurrent users

**AlgoVerse AI**

Sep. 2025 - Jun. 2026

LLM Research Intern

San Francisco, CA, USA (Remote)

- Engineered DeployStega, an LLM steganographic communication framework using token-binning, achieving 90%+ word inclusion fidelity validated against a 13M-event corpus spanning 695K+ real GitHub users
- Derived statistical priors from GH Archive to constrain synthetic trace generation within an empirically normal behavioral envelope
- Evaluated detection resistance against 5 adversarial classifiers (LR, RF, SVM, TS-CNN, BERT-base) on 4,000 synthetic users; adapted a differential-privacy  $\epsilon$  metric to quantify population-level indistinguishability, achieving sub-1.173  $\epsilon$

**DigiAlert**

Jun. 2025 - Aug. 2025

Software Engineer Intern

Chennai, TN, India

- Built a Flask-based AWS CSPM dashboard aggregating security metrics across 4 AWS services with Chart.js visualizations, cutting CSPM tooling costs by 45%
- Built a concurrent data pipeline (boto3, threading) processing 500+ CloudTrail events per refresh with real-time CIS v5.0.0 risk classification, reducing pipeline execution time by 35% vs. sequential baseline
- Deployed dashboard via nginx/gunicorn with REST API integration and TTL-based cache invalidation

## PROJECTS

**CUDA-Accelerated Order Book Reconstructor** | *CUDA, C++, CUB/Thrust, Nsight Systems*

- Built a CUDA order-book engine end-to-end with a custom ITCH 5.0 binary decoder, per-symbol stream compaction, and warp-per-symbol reconstruction, processing 305M+ NASDAQ messages at 4.4M msgs/sec across 9,000 symbols on an H100
- Swept 10,000 backtest configurations in parallel, compressing 6.5 hours of sequential backtesting into 83.8s for a 279 $\times$  speedup and 156 $\times$  end-to-end over a single-threaded C++ baseline on identical hardware
- Engineered a from-scratch GPU open-addressing hash table sustaining 99.99997% reconstruction accuracy, and eliminated 27 GB tick buffer via atomic-claim allocation over pinned SoA buffers

**SideQuest** | *JavaScript, CSS, HTML*

- Architected a production-ready Manifest V3 Chrome extension, structuring 9 modules across 3 isolated browser execution contexts (content scripts, service worker, popup) that communicate through a custom message-passing RPC layer
- Removed 100% of user setup friction (0 API keys, 0 sign-up, 0 configuration) by integrating with Claude.ai's web API to authenticate through the user's existing browser session, streaming responses via Server-Sent Events and auto-injecting surrounding conversation context so answers stayed on-topic
- Delivered a zero-backend, local-first design that passed a full security audit with 0 exposed credentials and 0 third-party data sharing, hardening the render path with HTML-escaping of all model output to eliminate XSS from untrusted content

## SKILLS

**Languages:** C++; C; Python; JavaScript; SQL; Verilog; MIPS Assembly

**Libraries and Frameworks:** CUDA; PyTorch; FastAPI; Flask; pandas; scikit-learn; NumPy; Matplotlib

**Tools:** NVIDIA Nsight; AWS; Azure; GCP; Git; TeX/LaTeX; Docker; Valgrind, Mathematica, MATLAB

**Other Activities:** CS 124 Project Manager; CodePath; [TEDx Speaker](#); ACM Corporate Team Member